ПРОИЗВОДНЫЕ. ВЫВОД.

I. $\frac{d}{dx} x^k = k \cdot x^{k-1}$

$$\frac{d}{dx} x^k = \lim_{\Delta x \to 0} \frac{(x+\Delta x)^k - x^k}{\Delta x} \ominus \left[(x+\Delta x)^k = x^k + k x^{k-1} \Delta x + \ldots \right] \ominus \lim_{\Delta x \to 0} \frac{x^k + k \cdot x^{k-1} \Delta x + \ldots - x^k}{\Delta x} = \lim_{\Delta x \to 0} k \cdot x^{k-1} + \ldots = k \cdot x^{k-1}$$

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II. $$\frac{d}{dx} \sin x = \lim_{\Delta x \to 0} \frac{\sin(x+\Delta x) - \sin x}{\Delta x} = \lim_{\Delta x \to 0} \frac{\sin x \cos \Delta x + \sin \Delta x \cos x - \sin x}{\Delta x} = \lim_{\Delta x \to 0} \frac{\sin x (1 - \cos \Delta x) + \sin \Delta x \cos x}{\Delta x} =$$

$$= \lim_{\Delta x \to 0} \sin x \left(\frac{1 - \cos \Delta x}{\Delta x} \right) + \cos x \left(\frac{\sin \Delta x}{\Delta x} \right) = \cos x$$

$\lim_{x \to 0} \frac{1 - \cos x}{x} = 0$

$\lim_{x \to 0} \frac{\sin \Delta x}{\Delta x} = 1$

III. $$\frac{d}{dx} \cos x = \lim_{\Delta x \to 0} \frac{\cos(x+\Delta x) - \cos x}{\Delta x} = \lim_{\Delta x \to 0} \frac{\cos x \cos \Delta x - \sin x \sin \Delta x - \cos x}{\Delta x} = \lim_{\Delta x \to 0} \cos x \left(\frac{\cos \Delta x - 1}{\Delta x} \right) - \sin x \left(\frac{\sin \Delta x}{\Delta x} \right) =$$

$$= - \sin x$$